1.) Walk through the derivation of u_{xx} in terms of (r,θ) and explain the steps.

- First, let's define $\frac{\partial f}{\partial x}$ and $\frac{\partial^2 f}{\partial x^2}$.

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial f}{\partial \theta} \cdot \frac{\partial \theta}{\partial x}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial r^2} \cdot \left(\frac{\partial r}{\partial x}\right)^2 + \frac{\partial^2 f}{\partial r \partial \theta}\left(\frac{\partial r}{\partial x} \cdot \frac{\partial \theta}{\partial x}\right)$$

$$+ \frac{\partial f}{\partial r} \cdot \frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 f}{\partial \theta \partial r}\left(\frac{\partial \theta}{\partial x} \cdot \frac{\partial r}{\partial x}\right)$$

$$+ \frac{\partial^2 f}{\partial \theta}\left(\frac{\partial \theta}{\partial x}\right)^2 + \frac{\partial f}{\partial \theta} \cdot \frac{\partial^2 \theta}{\partial x^2}$$

- With these equations, let's now define Polar Coordinates:

$x = r\cos\theta \quad y = r\sin\theta \quad r = \sqrt{x^2+y^2} \quad \theta = \arctan(y/x)$

- We can now take the various first and second derivatives of these functions.

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} = \frac{r\cos\theta}{r} = \cos\theta$$

$$\frac{\partial^2 r}{\partial x^2} = \frac{1}{\sqrt{x^2+y^2}} - \frac{x^2}{\sqrt{(x^2+y^2)^3}} = \frac{1}{r} - \frac{r^2\cos^2\theta}{r^3} = \frac{1}{r} - \frac{\cos^2\theta}{r} = \frac{\sin^2\theta}{r}$$

$$\frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2+y^2}} = \frac{r\sin\theta}{r} = \sin\theta$$

$$\frac{\partial^2 r}{\partial y^2} = \frac{1}{\sqrt{x^2+y^2}} - \frac{y^2}{\sqrt{(x^2+y^2)^3}} = \frac{1}{r} - \frac{r^2\sin\theta}{r^3} = \frac{1}{r} - \frac{\sin^2\theta}{r} = \frac{\cos^2\theta}{r}$$

Cont...

$$\frac{\partial^2 r}{\partial x \partial y} = \frac{xy}{(x^2+y^2)^{3/2}} = \frac{r^2 \cos\theta \sin\theta}{r^3} = \frac{\cos\theta \sin\theta}{r} = \frac{\partial^2 r}{\partial y \partial x}$$ (mixed partials equal)

$$\frac{\partial \theta}{\partial x} = \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \frac{-y}{x^2} = \frac{-y}{x^2+y^2} = \frac{-r\sin\theta}{r^2} = \frac{-\sin\theta}{r}$$

$$\frac{\partial^2 \theta}{\partial x^2} = \frac{2yx}{(x^2+y^2)^2} = \frac{2r^2 \sin\theta \cos\theta}{r^4} = \frac{2\sin\theta\cos\theta}{r^2}$$

$$\frac{\partial \theta}{\partial y} = \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \frac{1}{x} = \frac{x}{x^2+y^2} = \frac{r\cos\theta}{r^2} = \frac{\cos\theta}{r}$$

$$\frac{\partial^2 \theta}{\partial y^2} = \frac{-x \cdot 2y}{(x^2+y^2)^2} = \frac{-2r^2\cos\theta\sin\theta}{r^4} = \frac{-2\cos\theta\sin\theta}{r^2}$$

$$\frac{\partial^2 \theta}{\partial x \partial y} = \frac{1}{x^2+y^2} - \frac{x \cdot 2x}{(x^2+y^2)^2} = \frac{1}{r^2} - \frac{2r^2\cos^2\theta}{r^4} = \frac{\sin^2\theta - \cos^2\theta}{r^2} = \frac{\partial^2 \theta}{\partial y \partial x}$$

- We can rewrite $\frac{\partial f}{\partial x}$ as follows: $u_x = u_r \cdot r_x + u_\theta \cdot \theta_x$
- We can then define $\frac{\partial}{\partial x} \cdot \frac{\partial f}{\partial x}$ as follows:

$$\frac{\partial}{\partial x} \cdot \frac{\partial f}{\partial x} = \frac{\partial^2 f}{\partial x^2} = u_{xx} = \frac{\partial}{\partial x}\left[u_r \cdot r_x + u_\theta \cdot \theta_x\right] = u_{rr}(r_x)^2 + u_r r_{xx}$$

$$+ u_{\theta\theta}(\theta_x)^2 + u_\theta \theta_{xx}$$

- Now we just have to sub. in the various derivatives we found above.

cont...

$$u_{rr}(r_x)^2 + u_r r_{xx} + u_{\theta\theta}(\theta_x)^2 + u_\theta \theta_{xx}$$

$$= u_{rr}[\cos\theta]^2 + u_r\left[\frac{\sin^2\theta}{r}\right] + u_{\theta\theta}\left[\frac{-\sin\theta}{r}\right] + u_\theta\left[\frac{2\sin\theta\cos\theta}{r^2}\right]$$

- This is our final answer. Q.E.D.

2.) Apply a separation of variables in (r, θ) to define the ODEs for the given PDE on a disk: $u_y + a u_x = 0$

- We previously defined u_x as follows: $u_x = u_r r_x + u_\theta \theta_x$

- We can also expand this to u_y: $u_y = u_r r_y + u_\theta \theta_y$

- Let's now substitute these values into the original eq'n:

$$[u_r r_y + u_\theta \theta_y] + a[u_r r_x + u_\theta \theta_x] = 0$$

- We know $u(r,t) = R(r)T(t)$, and that we defined the various "r" derivatives in the previous problem. (Same with "θ" derivatives). Let's now substitute the "r" + "θ" derivatives:

$$\left\{u_r[\sin\theta] + u_\theta\left[\frac{\cos\theta}{r}\right]\right\} + a\left[u_r[\cos\theta] + u_\theta\left[\frac{-\sin\theta}{r}\right]\right] = 0$$

$$\rightarrow u_r \sin\theta + u_\theta \frac{\cos\theta}{r} + a u_r \cos\theta - a u_\theta \frac{\sin\theta}{r} = 0$$

$$\rightarrow u_r \sin\theta + a u_r \cos\theta + u_\theta \frac{\cos\theta}{r} - a u_\theta \frac{\sin\theta}{r} = 0$$

$$\rightarrow u_r(\sin\theta + a\cos\theta) + u_\theta\left(\frac{\cos\theta}{r} - \frac{a\sin\theta}{r}\right) = 0$$

$$\rightarrow u_r(\sin\theta + a\cos\theta) = -u_\theta\left(\frac{\cos\theta - a\sin\theta}{r}\right)$$

- We can now invoke $u(r,\theta) = R(r)\Theta(\theta)$ and SOV to define the ODE

$$\left.\begin{aligned} u(r,\theta) &= R(r)\Theta(\theta) \\ u_r(r,\theta) &= R'\Theta \\ u_\theta(r,\theta) &= R\Theta' \end{aligned}\right\} \xrightarrow{\text{Sub into eq'n}} R'\Theta(\sin\theta + a\cos\theta) = -R\Theta'\left(\frac{\cos\theta - a\sin\theta}{r}\right)$$

$(\div$ by $R\Theta)$

$$\rightarrow \frac{R'}{R}(\sin\theta + a\cos\theta) = \frac{-\Theta'}{\Theta}\left(\frac{\cos\theta - a\sin\theta}{r}\right)$$

Cont...

- We left off with the following:

$$\frac{R'}{R}(\sin\theta + a\cos\theta) = \frac{-\theta'}{\theta}\left(\frac{\cos\theta - a\sin\theta}{r}\right) \quad (\div \text{ by } \sin\theta + a\cos\theta)$$

$$\rightarrow \frac{R'}{R} = \frac{-\theta'}{\theta}\left(\frac{\cos\theta - a\sin\theta}{r(\sin\theta + a\cos\theta)}\right) \quad \left(\text{define } \left[\frac{\cos\theta - a\sin\theta}{r(\sin\theta + a\cos\theta)}\right] \text{ as } \frac{1}{c^2}\right)$$

$$\rightarrow \frac{R'}{R} = \frac{-\theta'}{c^2\theta} = -\lambda^2 \left\} \begin{array}{l} R' + \lambda^2 R = 0 \\ \theta' - \lambda^2 c^2 \theta = 0 \end{array}\right.$$

- These are our final answers.

Q.E.D.

3.) Solve the axisymmetric wave equation on a disk for the following values:

$u_{tt} = c^2\left[u_{rr} + \frac{1}{r}u_r\right]$ for $0<r<2$, $u(r,0) = 1-r^2$, $u_t(r,0) = 1$

- Since $u(r,t) = R(r)T(t)$, take derivs. + sub.

$R(r)T''(t) = c^2\left[R''(r)T(t) + \frac{1}{r}R'(r)T(t)\right]$ ($\div$ by c^2RT):

$$\frac{T''}{c^2T} = \frac{R''}{R} + \frac{R'}{rR} = -\lambda^2 \quad \text{(form ODEs)}$$

$$\Rightarrow \left.\begin{array}{l} T'' + c^2\lambda^2 T = 0 \\ R'' + \frac{R'}{r} + \lambda^2 R = 0 \end{array}\right\}$$ while the 2nd eq'n requires Bessel function, the first can be solved using previous means; lets do that.

$T'' + c^2\lambda^2 T = 0 \rightarrow r^2 + c^2\lambda^2 = 0 \rightarrow r = \pm c\lambda i \rightarrow T(t) = E_1\cos(c\lambda t) + E_2\sin(c\lambda t)$

- We know the following about Bessel functions:

$$J_p(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k+p+1)}\left(\frac{x}{2}\right)^{2k+p} \Big/ J_{-p}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k-p+1)}\left(\frac{x}{2}\right)^{2k-p}$$

Second solution: $$Y_p(x) = \frac{J_p(x)\cos(p\pi) - J_{-p}(x)}{\sin(p\pi)} \quad p \in \mathbb{Z}$$

- General sol'ns: $y(x) = c_1 J_p(x) + c_2 Y_p(x)$ (Bessel)

- General radially symmetric wave eq'n sol'ns:

$$u(r,t) = \sum_{n=1}^{\infty}\left[A_n\cos(c\lambda_n t) + B_n\sin(c\lambda_n t)\right]J_0(\lambda_n r) \quad \text{where}$$

$$A_n = \frac{2}{a^2 J_1^2(\alpha_n)}\int_0^a f(r)J_0(\lambda_n r)\,r\,dr$$

$$B_n = \frac{2}{c\alpha_n a J_1^2(\alpha_n)}\int_0^a g(r)J_0(\lambda_n r)\,r\,dr$$

$\lambda_n = \frac{\alpha_n}{a}$, $\alpha_n = n^{th}$ positive zero of J_0

$$J_0(\lambda_n X) = \sum_{K=0}^{\infty} \frac{(-1)^K}{K!\,\Gamma(K+1)} \left(\frac{\lambda_n X}{2}\right)^{2K} \quad (\text{let } p=0 + X=\lambda_n X)$$

$$A_n = \frac{2}{a^2 J_1^2(\alpha_n)} \int_0^a [1-r^2] \left[\sum_{K=0}^{\infty} \frac{(-1)^K}{K!\,\Gamma(K+1)} \left(\frac{\lambda_n r}{2}\right)^{2K}\right] r\,dr$$

$$B_n = \frac{2}{\alpha_n \cdot c \cdot a \cdot J_1^2(\alpha_n)} \int_0^a \left[\sum_{K=0}^{\infty} \frac{(-1)^K}{K!\,\Gamma(K+1)} \left(\frac{\lambda_n r}{2}\right)^{2K}\right] r\,dr$$

$\lambda_n = \frac{\alpha_n}{a}$ where α_n is the nth positive zero of J_0

Q.E.D.

4.) Solve Laplace's Eq'n on a disk for the following values:

$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0$ for $0<r<3$, $0<\theta<2\pi$, and $u(3,\theta) = \cos(3\theta)$

- seek a sol'n to $u(r,\theta) = R(r)\,\Theta(\theta)$ and apply SoV:

$$R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta'' = 0 \qquad (* \text{ by } r^2;\ \div \text{ by } R\Theta)$$

$$\frac{r^2R''}{R} + \frac{rR'}{R} + \frac{\Theta''}{\Theta} = 0 \rightarrow \frac{r^2R''}{R} + \frac{rR'}{R} = -\frac{\Theta''}{\Theta} = \lambda \left\{ \begin{array}{l} r^2R'' + rR' - \lambda R = 0 \\ \Theta'' = -\lambda\Theta \end{array} \right.$$

$\rightarrow \Theta_n(\theta) = C_n\cos(n\theta) + D_n\sin(n\theta)$; $\lambda = n^2$

$\rightarrow r^2R'' + rR' - n^2R = 0 \rightarrow$ Euler's Eq'n w/ sol'ns $R(r) = C_1\left(\frac{r}{a}\right)^n + C_2\left(\frac{r}{a}\right)^{-n}$

$\cdot\, C_2 = 0$ as $r \rightarrow 0$

$$\Rightarrow u(r,\theta) = a_0 + \sum_{n=1}^{\infty}\left(\frac{r}{a}\right)^n\left[A_n\cos(n\theta) + B_n\sin(n\theta)\right]$$

$$u(3,\theta) = a_0 + \sum_{n=1}^{\infty} A_n\cos(n\theta) + B_n\sin(n\theta) = \cos(3\theta)$$

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi}\cos(3\theta)\,d\theta = \frac{1}{2\pi}\cdot\frac{1}{3}\sin(3\theta)\Big|_0^{2\pi} = \frac{1}{6\pi}\left\{[\sin(2\pi)] - [\sin(0)]\right\} = 0$$

$$A_n = \frac{1}{\pi}\int_0^{2\pi}\cos(3\theta)\cos(n\theta)\,d\theta = \begin{cases} 1 & \text{if } n=3 \\ 0 & \text{otherwise} \end{cases}$$

$$B_n = \frac{1}{\pi}\int_0^{2\pi}\cos(3\theta)\sin(n\theta)\,d\theta = 0$$

$$u(r,\theta) = \left(\frac{r}{3}\right)\cos(3\theta)$$

Q.E.D.

5.) Solve Laplace's Equation on a disk for the following values:

$u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 0$ for $0 < r < 5$, $0 < \theta < 2\pi$, + $u(5,\theta) = 1-\theta$

$$u(5,\theta) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\theta) + D_n \sin(n\theta) = 1-\theta$$

$$c_0 = \frac{1}{2\pi}\int_0^{2\pi} (1-\theta)\,d\theta = \frac{1}{2\pi}\left\{\theta - \frac{\theta^2}{2}\right\}\Big|_0^{2\pi} = \frac{1}{2\pi}\left\{[2\pi - 2\pi^2] - [0-0]\right\}$$

$$= \frac{2\pi - 2\pi^2}{2\pi}$$

$$c_n = \frac{1}{\pi}\int_0^{2\pi} (1-\theta)\cos(n\theta)\,d\theta \quad \left\{\begin{array}{ccc|c} + & 1-\theta & & \cos(n\theta) \\ - & -1 & & \frac{1}{n}\sin(n\theta) \\ + & 0 & & -\frac{1}{n^2}\cos(n\theta) \end{array}\right.$$

$$= \frac{(1-\theta)}{n}\sin(n\theta) - \frac{1}{n^2}\cos(n\theta)\Big|_{\theta=0}^{\theta=2\pi} = \left\{\left[\frac{1-2\pi}{n}\sin(2n\pi) - \frac{\cos(2n\pi)}{n^2}\right] - \left[0 - \frac{\cos(0)}{n^2}\right]\right\}$$

$$= \frac{-(+1)^n}{n^2} + \frac{1}{n^2} = \frac{1-(+1)^n}{n^2} \qquad c_n = \frac{1}{\pi}\left[\frac{1-(+1)^n}{n^2}\right] = 0$$

$$D_n = \frac{1}{\pi}\int_0^{2\pi} (1-\theta)\sin(n\theta)\,d\theta \quad \left\{\begin{array}{ccc|c} + & 1-\theta & & \sin(n\theta) \\ - & -1 & & -\frac{1}{n}\cos(n\theta) \\ + & 0 & & -\frac{1}{n^2}\sin(n\theta) \end{array}\right.$$

$$-\frac{(1-\theta)}{n}\cos(n\theta) - \frac{1}{n^2}\sin(n\theta)\Big|_{\theta=0}^{\theta=2\pi} \quad \left\{\left[\frac{2\pi-1}{n}\cos(2n\pi) - 0\right] - \left[\frac{-1}{n}\cos(0) - 0\right]\right\}$$

$$= \frac{2\pi-1}{n}(1)^n + \frac{1}{n} = \frac{2\pi}{n} \qquad D_n = \frac{1}{\pi}\cdot\frac{2\pi}{n} = \frac{2}{n}$$

$$u(r,\theta) = \frac{2\pi - 2\pi^2}{2\pi} + \sum_{n=1}^{\infty}\left(\frac{r}{5}\right)^n \cdot \frac{2}{n}\cdot\sin(n\theta)$$

Q.E.D.

Homework 11

Adam Prieto
Dr. Shelley Poole
MTH 3440 - Partial Differential Equations
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Chapter 4 Problems

6.) Discuss the application of separation of variables in Polar and Cylindrical problems. How does it relate to the work you did in rectangular coordinates? What do you find most striking about the solution forms?

Although the general process remains the same, separation of variables in Polar and Cylindrical coordinates is significantly harder than in rectangular. This is due to the fact that the definition of the derivatives in polar and cylindrical require many partial derivatives (and work) when deriving the desired formulas. Then comes substitution. This problem will become more apparent when discussing spherical coordinates, but for now just know that the algebra gets pretty messy and very complex.

In regards to the resulting ODEs, aside from the different variables, the equations themselves are identical to their rectangular counterparts, and can be solved as such. Problem 2 from chapter 4 produces the familiar $R' + \lambda^2 R = 0$ equation that we've seen countless times already in this class, so we can get a solution fairly easily (check the previous work for the derivation). However, it's important to note that not all the solutions are *identical* to the ones we seen before, since I had to manipulate the $\Theta(\theta)$ equation; but, I was still able to get something reminiscent of what I've worked with before.

Chapter 5 Problems

1.)Walk through the derivation of u_{xx} in terms of

(r, θ, ϕ) and explain the steps.

When defining u_{xx} in terms of (r, θ, ϕ), we should first apply the chain rule as we did in polar coordinates. Doing so yields the following:

$u_{xx} = u_{\rho\rho}(\rho_x)^2 + u_{\rho\theta}\,\rho_x\,\theta_x + u_{\rho\phi}\,\rho_x\,\phi_x + u_\rho\,\rho_{xx} + u_{\theta\theta}(\theta_x)^2 +$ (←This can also be found in

$u_{\theta\rho}\,\theta_x\,\rho_x + u_{\theta\phi}\,\theta_x\,\phi_x + u_\theta\,\theta_{xx} + u_{\phi\phi}(\phi_x)^2 + u_{\phi\rho}\,\phi_x\,\rho_x + u_{\phi\theta}\,\phi_x\,\theta_x + u_\phi\,\phi_{xx}$.

the notes.)

Obviously, this equation is quite a bit to work with, and is made easier with Mathematica. That said, lets now define ρ, θ, and ϕ in terms of the variables x, y, and z.

In[1]:=
```
ρ[x_, y_, z_] = Sqrt[x^2 + y^2 + z^2];
θ[x_, y_, z_] = ArcTan[Sqrt[x^2 + y^2] / z];
ϕ[x_, y_, z_] = ArcTan[y / x];
```

The nice thing about Mathematica is that we can take infinitely many derivatives by only changing a few characters on a line. However, for our purposes, we only need the first and second derivatives of the above functions:

In[4]:=
```
a = FullSimplify[D[ρ[x, y, z], {x, 1}]] (*ρx*)
b = FullSimplify[D[ρ[x, y, z], {x, 2}]](*ρxx*)
c = FullSimplify[D[θ[x, y, z], {x, 1}]](*θx*)
d = FullSimplify[D[θ[x, y, z], {x, 2}]](*θxx*)
e = FullSimplify[D[ϕ[x, y, z], {x, 1}]](*ϕx*)
f = FullSimplify[D[ϕ[x, y, z], {x, 2}]](*ϕxx*)
```

Out[4]= $\dfrac{x}{\sqrt{x^2 + y^2 + z^2}}$

Out[5]= $\dfrac{y^2 + z^2}{\left(x^2 + y^2 + z^2\right)^{3/2}}$

Out[6]= $\dfrac{x\,z}{\sqrt{x^2 + y^2}\,\left(x^2 + y^2 + z^2\right)}$

Out[7]= $\dfrac{z\,\left(-2\,x^4 - x^2\,y^2 + y^4 + y^2\,z^2\right)}{\left(x^2 + y^2\right)^{3/2}\,\left(x^2 + y^2 + z^2\right)^2}$

Out[8]= $-\dfrac{y}{x^2 + y^2}$

Out[9]= $\dfrac{2\,x\,y}{\left(x^2 + y^2\right)^2}$

Now we can combine all of the above results and get the derivation for u_{xx} in terms of (r, θ, ϕ):

In[10]:=
$$u_{xx} = u_{\rho\rho}\,(a)^2 + u_{\rho\theta}\,(a*c) + u_{\rho\phi}\,(a*e) + u_{\rho}\,(b) + u_{\theta\theta}\,(c)^2 + u_{\theta\rho}\,(c*a) + u_{\theta\phi}\,(c*e) + u_{\theta}\,(b) + u_{\phi\phi}\,(e)^2 + u_{\phi\rho}\,(e*a) + u_{\phi\theta}\,(e*c) + u_{\phi}\,(f)$$

Out[10]=
$$\frac{(y^2+z^2)\,u_\theta}{(x^2+y^2+z^2)^{3/2}} + \frac{x^2 z^2 u_{\theta\theta}}{(x^2+y^2)\,(x^2+y^2+z^2)^2} + \frac{x^2 z\, u_{\theta\rho}}{\sqrt{x^2+y^2}\,(x^2+y^2+z^2)^{3/2}} - \frac{x\,y\,z\,u_{\theta\phi}}{(x^2+y^2)^{3/2}\,(x^2+y^2+z^2)} + \frac{(y^2+z^2)\,u_\rho}{(x^2+y^2+z^2)^{3/2}} + \frac{x^2 z\, u_{\rho\theta}}{\sqrt{x^2+y^2}\,(x^2+y^2+z^2)^{3/2}} + \frac{x^2 u_{\rho\rho}}{x^2+y^2+z^2} - \frac{x\,y\,u_{\rho\phi}}{(x^2+y^2)\,\sqrt{x^2+y^2+z^2}} + \frac{2\,x\,y\,u_\phi}{(x^2+y^2)^2} - \frac{x\,y\,z\,u_{\phi\theta}}{(x^2+y^2)^{3/2}\,(x^2+y^2+z^2)} - \frac{x\,y\,u_{\phi\rho}}{(x^2+y^2)\,\sqrt{x^2+y^2+z^2}} + \frac{y^2 u_{\phi\phi}}{(x^2+y^2)^2}$$

Obviously this answer is a lot to work with, but putting it in a Mathematica document greatly simplifies the grunt work. Imagine having to work this out by hand.

2.) Discuss the forms found in Spherical coordinates—both of the ODEs and their solutions. Do some research into Legendre's differential equation, functions, and the resulting spherical harmonics. Summarize your takeaways and cite all references.

Much like our previous solutions in polar, our current solutions in spherical produce long, complicated solutions with lots of trig functions. It's also worth noting that many more partial derivatives need to be taken since we're dealing with a higher dimension. Tediousness aside, we can begin analyzing the results. To my surprise, the ODE for the azimuthal angle—$\Phi'' + \lambda * \Phi = 0$—is the exact same ODE that we've solved countless times in this class, and can do so once again right here. Even the equation for R—$\rho^2 R'' + 2\rho R' - \mu R = 0$—isn't too dissimilar; this particular Euler equation can be solved using previous ODE techniques and is even similar to what we saw in polar. Θ, however, is another story. Converting the Θ equation requires a change of variables which is tedious in itself. For brevity sake, I'm just going to list the final answer here: $(\mathrm{Sin}[\theta])^2 * \Theta_{ss} - 2\,\mathrm{Cos}[\theta] * \Theta_s + \left(\mu - \frac{m^2}{(\mathrm{Sin}[\Theta])^2}\right) * \Theta = 0$. I personally didn't recognize this equation at first, but was able to figure out what was going on after reading the provided notes. To quote the text directly:

This is actually an associated Legendre differential equation, as it doesn't exactly match the form

above. In the symmetric case, the last term is simplified to match the form of Legendre's differential equation.

For these solutions, we define the solutions in terms of m and μ=n(n+1) through the associated Legendre functions denoted $(P^m)_n$ Cos[θ].

-(*Partial Differential Equations-An Exploration in Modeling Our Physical World*, Dr. Shelley Poole, 118)

This begs the question: what is a Legendre Polynomial? Time for some outside sources.

When researching Legendre's differential equation, my first stop was to the assigned textbook for our class. As it turns out, the textbook is extremely detailed on Legendre Polynomials and related topics. For brevity sakes, I'm going to reference the text exclusively from here on out. That said, let's talk about what it has to offer.

The text goes into great detail about solving the equation using power series, and also mentions how the series will converge for |x| < 1. We can use techniques that we learned back in ODEs to accomplish this task, so there's no new math here. However, it's important to point out that several relations arise from this approach, such as Bonnet's (again, this is discussed in the book). In regards to spherical harmonics, these functions arise when working on the surface of a sphere, and so these phenomena are going to pop up here. The text gives a formal definition on page 284, so I'm just going to refer you to that page for details as well as the following pages for how spherical harmonics can be used to solve particular PDEs. In all, these are interesting fields of study that may pop up again in higher level math classes.